

Defense Q&A Preparation Guide

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Companion to `defense_presentation.pdf`

This guide closes the only gap that a slide deck cannot address: delivery style and committee Q&A. It is a private preparation document for the candidate; it is not shown to the committee.

1 The One-Sentence Identity

When anyone asks “What is your dissertation about?” — regardless of how the question is phrased — the answer is always the same single sentence:

This dissertation investigates how Responsible AI governance mechanisms improve clinician trust and operational adoption readiness in AI-assisted healthcare environments.

Use this answer verbatim in:

- Opening introduction
- “What is your dissertation about?”
- “Is this governance / adoption / explainability / healthcare AI research?”
- “Can you summarize your contribution in one sentence?”
- Closing remarks

Repeating one locked sentence is the supervisor’s specific instruction. Improvisation under pressure introduces drift toward technical or architecture framing — the failure mode this entire dissertation reframing was built to prevent.

2 Delivery Style Rules (the #17 gap)

2.1 Speak slower, simpler, more organizationally

The candidate’s natural tendency is to think *deep, broad, multi-layered*. During defense, this overwhelms committee. Apply five delivery rules:

1. **One idea per sentence.** If you need a semicolon, you are overpacking.
2. **Pause for two beats after each finding.** The committee needs processing time; you do not.
3. **Replace “the architecture” with “the framework”** every time the word comes up. Architecture invites engineering questions; framework invites contribution questions.
4. **Replace “performance” with “adoption”** when discussing results. “The classifier performed at 97%” invites accuracy debate; “the framework supports adoption because trust forms when explainability is visible” invites contribution discussion.
5. **When asked an engineering question, redirect to governance.** Example: “How did you tune the model?” → “The tuning details are in Appendix C; what matters for this dissertation is that the model’s output was visible to clinicians through the explainability layer, which is what produced the trust effect.”

2.2 Language to use; language to avoid

Use deliberately:

- “Governance-driven explainability”
- “Clinician trust”
- “Operational adoption readiness”
- “Calibrated trust” (not absolute trust)
- “Decision support” (not autonomous diagnosis)
- “Human-centered Responsible AI governance”
- “Framework” (not platform / architecture / system)

Never say (these weaken DBA credibility):

- “Advanced AI architecture”
- “Multi-agent system”
- “AGI”, “quantum AI”, “swarm AI”, “neuromorphic AI”, “digital twin”
- “Autonomous diagnosis”
- “Enterprise transformation”
- “Platform”
- “Pipeline” (use “workflow” instead)
- “Production-ready” (use “operationally deployable” instead)

3 Top 25 Anticipated Questions with One-Sentence Answers

Each question has a single locked answer. Memorize these. They are designed to be defensible, not impressive.

3.1 Framing questions

Question	Locked answer
What is your dissertation about?	Investigates how Responsible AI governance mechanisms improve clinician trust and operational adoption readiness in AI-assisted healthcare environments.
Is this governance research or adoption research?	Both: governance is the antecedent, adoption is the outcome, trust is the mediator.
Is this an AI architecture dissertation?	No: the dissertation is human-centered Responsible AI governance research; the architecture exists in the appendix as supporting operational context, not as contribution.
What is the central research question?	How does Responsible AI governance influence clinician trust and adoption intention for AI-assisted ASD-EEG screening, through perceived explainability?
Why ASD-EEG specifically?	Six reasons — early-intervention sensitivity, diagnostic subjectivity, trust sensitivity, regulatory high-risk classification, decision-support fit, and paediatric ethical sensitivity — which together make ASD-EEG the most appropriate empirical context.

3.2 Theory & methodology questions

Question	Locked answer
Why TAM and not UTAUT or DeLone-McLean?	TAM was extended by positioning Responsible AI governance as a structural antecedent of trust; UTAUT conflates governance with facilitating conditions, and DeLone-McLean treats quality as unitary.
Why pragmatism as the research philosophy?	The research question requires both quantitative measurement (effect sizes) and qualitative interpretation (why clinicians do or do not adopt); neither paradigm alone answers both.
Why PLS-SEM rather than CB-SEM?	PLS-SEM is recommended for $n < 200$, exploratory-predictive research, and complex mediation structures; all three apply here.
How do you justify $n = 131$?	Three grounds: Hair 10:1 rule ($n_{\min} = 100$), G^* Power analysis ($n_{\min} = 119$), and Kock-Hadaya healthcare PLS-SEM norms ($n = 80\text{--}150$).
What is your mixed-methods design?	Convergent parallel design after Creswell, with the joint-display matrix in Chapter 4 as the integration mechanism.

3.3 Construct & measurement questions

Question	Locked answer
Is explainability just another form of trust?	No: Fornell-Larcker AVE diagonals exceed off-diagonal correlations, HTMT is 0.78 (below 0.85 threshold), and the constructs load on distinct factors in CFA.
Where does the trust measurement come from?	Madsen & Gregor (2000) Trust in Automation Index (TUI), 10 items, adapted from “automation” to “AI system” terminology.
Where does the governance measurement come from?	Adapted from EU AI Act (2024) control taxonomy plus Floridi et al. (2018) principles, operationalized as 12 items across seven control families.
How did you handle common method bias?	Harman’s single-factor test (first factor 32.8%, well below 50% threshold), marker-variable technique (correlations < 0.20), and procedural controls (anonymity, reverse-coded items, temporal separation).
Did you address non-response bias?	Yes: Armstrong-Overton (1977) early-vs-late respondent comparison shows no significant differences on any construct ($p > .05$).

3.4 Findings & interpretation questions

Question	Locked answer
Which hypotheses were supported?	H1–H5 fully supported; H6 partially supported (moderation effect only for low-AI-familiarity clinicians).
What is the effect size on trust?	The Explainability → Trust path coefficient is $\beta = 0.71$, $p < .001$ — a strong effect by Cohen’s conventional thresholds.
Why is the mediation effect significant?	The bootstrap 95% CI for the indirect effect excludes zero, and the Sobel test is significant ($z = 4.21$, $p < .001$); H4 is full mediation, H5 is partial mediation.
Are you not over-claiming causation?	No: the findings establish statistical associations consistent with the hypothesised pathway, not deterministic causation; this is explicitly stated in Chapter 4.
What about negative findings?	Five negative findings are reported in Chapter 4: over-reliance concern (31% of clinicians), explainability-alone insufficiency, workflow-disruption fear, governance maturity variability, and partial moderation.

3.5 Contribution & scope questions

Question	Locked answer
What is your theoretical contribution?	Extends Technology Acceptance Model by positioning Responsible AI governance as a structural antecedent of trust → adoption in clinical AI, with sequential mediation specified.
What is your practical contribution?	The RGAIG Maturity Framework, a six-level managerial instrument for healthcare AI deployment readiness assessment.
Why didn’t you include [AGI/quantum/swarm/digital-twin]?	Scope discipline: these are speculative or outside the governance-trust-adoption pathway, and their inclusion would dilute the dissertation’s contribution focus.
What are the limitations?	Cross-sectional design, $n = 131$, ASD-only domain, self-reported intention rather than observed behaviour, geographic concentration, and absence of longitudinal panel data.
What is the future work?	Longitudinal governance evaluation, multi-site validation, additional neurological domains, real-world deployment, comparative maturity benchmarking, and clinician-AI collaboration studies.

4 Q&A Handling Strategy

4.1 When asked a question outside scope

Pattern: “[Off-scope topic] is genuinely interesting and is documented in [Appendix X / future work]. For this dissertation, the relevant construct is [governance / explainability / trust / adoption], which is what the [hypothesis number] result addresses.”

Examples:

- “Could you use a transformer instead of CNN?” → “Architecture choice is documented in Appendix C; what matters here is that the model’s output was rendered explainable to clinicians, which is what produced the trust effect.”
- “What about quantum AI?” → “Quantum AI is explicitly excluded from scope (Appendix H Scope Control Pack); it is not relevant to the governance-trust-adoption pathway this dissertation tests.”
- “Could this scale to other diseases?” → “Cross-condition validation is named as future work; the dissertation tests the pathway in ASD-EEG specifically and states explicit boundary conditions for generalization.”

4.2 When asked a hostile or challenging question

Three-step pattern:

1. Acknowledge the question’s validity briefly (“That is a fair question.”)
2. Restate the question in your own words to confirm you understood
3. Answer with the locked one-sentence response, then offer one supporting detail

Examples:

- “Isn’t this just a TAM study?” → “That is a fair critique. The question is whether positioning RAI governance as a structural antecedent of TAM’s perceived-usefulness construct constitutes a meaningful extension. The dissertation argues yes, on three grounds: TAM presupposes value is observable; in safety-critical AI it is not; governance is the upstream variable that makes the perception possible.”
- “Is $n = 131$ enough?” → “A reasonable concern. The sample is justified on three independent grounds: Hair 10:1 rule, G^* Power for medium effect, and published norms for healthcare PLS-SEM. Limitations of this n are explicitly stated in the limitations section.”
- “How is this not just a survey?” → “The dissertation has four contribution types, not just empirical survey results. The RGAIG Maturity Framework is the managerial artefact, the validated instrument is the methodological contribution, the TAM extension is the theoretical contribution.”

4.3 When you do not know the answer

Never bluff. Use one of three responses:

1. “That is outside the scope of this dissertation; it is named as future work in [section].”
2. “I would need to verify that; can I follow up with you in writing?”
3. “The supporting evidence is in [Appendix X]; let me reference it.”

5 Slide-by-Slide Delivery Notes

5.1 Opening trilogy (slides 1–3)

These three slides establish the entire defense framing. If they go well, the rest of the defense is on autopilot. Three rules:

- **Slide 1 (title):** Pause after the title. Do not start speaking immediately. Let the committee read the title.
- **Slide 2 (Problem):** Start with “Healthcare organizations face significant barriers...” — not with “My research is about...”
- **Slide 3 (Research Gap):** Read the table column headers aloud. The 2-column structure does the framing work; do not over-explain.

5.2 Centerpiece slides (5, 9, 10)

Slide 5 (Conceptual Framework) is the most important slide.

- Read the four box labels in sequence: Responsible AI Governance, Explainability, Trust, Adoption.
- Then state the H1–H3 path in plain language: “Governance affects explainability; explainability affects trust; trust affects adoption.”
- Do not start with H4/H5/H6 mediation. Those are slide 7.

Slide 9 (RGAIG Maturity Framework) is the hero artefact.

- This slide should take 90 seconds, not 30.
- Walk through Levels 1–6 sequentially.
- Close with: “This is the dissertation’s primary practical contribution: a six-level managerial instrument that healthcare CIOs can apply directly.”

Slide 10 (Findings) is storytelling, not statistics dump.

- For each row, say the finding column FIRST, then the meaning column.
- Statistical detail (effect sizes, *p*-values) is in backup B1. Do not recite numbers from slide 10 unless asked.

5.3 Closing trilogy (slides 18–20)

These slides lock the dissertation’s identity. Three rules:

- **Slide 18 (Unified Identity):** Pause before clicking. Read the italic sentence aloud, then say nothing else. Let the committee absorb.
- **Slide 19 (Final Doctoral Claim):** Same pattern. Read the claim verbatim from the slide. Do not paraphrase.
- **Slide 20 (Thank You):** Brief: “Thank you. I welcome your questions.” Then sit.

6 Anti-Pattern Reminders (the things to NOT do)

- **Do not pull up architecture diagrams unprompted.** The architecture backup slide (B9) is only shown if the committee asks specifically about deployment, infrastructure, or system design.
- **Do not defend accuracy numbers.** If asked about 97% accuracy, redirect: “Accuracy is necessary but not sufficient; trust is the binding constraint, and that is what this dissertation measures.”
- **Do not introduce new concepts mid-defense.** Stick to constructs defined in slide 6.
- **Do not promise future deployments.** The framework is conceptual; production deployment is future work.
- **Do not name vendor products.** “SHAP” and “RAG” are method labels (acceptable); “LangChain” / “OpenAI” / “Databricks” weaken academic credibility.
- **Do not show emotion when challenged.** The supervisor critique cycle has hardened every claim. The defense is the easy part.

7 Pre-Defense Checklist (24 hours before)

- ☐ Reread `main_full.pdf` Section “Doctoral Research Framing Summary” (Chapter 1) and Section “Final Doctoral Claim” (Chapter 5)
- ☐ Memorize the one-sentence identity verbatim
- ☐ Memorize the locked final doctoral claim verbatim
- ☐ Run through slides 1–3 aloud, twice
- ☐ Run through slides 18–20 aloud, twice
- ☐ Confirm `defense_presentation.pdf` renders correctly on the presentation laptop
- ☐ Print 2 paper copies of the slide deck as backup
- ☐ Prepare 5 minutes of opening remarks (not reading from slides)
- ☐ Prepare 30 seconds of closing remarks
- ☐ Sleep

The dissertation is defensible. Thirteen Defensibility Packs, $n = 131$ empirical evidence, RGAIG Maturity Framework as the hero artefact, four theoretical anchors, one locked identity. The work is done. The defense is the easy part.

This Q&A Preparation Guide is a private candidate document. It is not part of the dissertation submission. Last updated alongside `defense_presentation.pdf`.